CS23532 - COMPUTER NETWORKS LABORATORY

EXPERIMENT 8

AIM:

To Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform (Room Link: https://tryhackme.com/room/nmap01).

DESCRIPTION:

- **Nmap Overview:** Nmap (Network Mapper) is a free, open-source tool created by Gordon Lyon (Fyodor) in 1997 for network discovery, inventory, vulnerability scanning, and security auditing.
- **Host Discovery Stage:** Before conducting port scans, Nmap identifies active/live systems to avoid wasting time and generating unnecessary network noise.
- **Discovery Techniques:**
    - **ARP Scan (-PR):** Uses ARP requests to discover live hosts on the same local subnet.
    - **ICMP Scan (-PE, -PP, -PM):** Uses ICMP Echo requests, Timestamp requests, and Address Mask requests to identify online systems when firewalls permit.
    - **TCP/UDP Ping Scans (-PS, -PA, -PU):** Sends packets to TCP (SYN/ACK) and UDP ports to determine live hosts when ICMP is blocked.

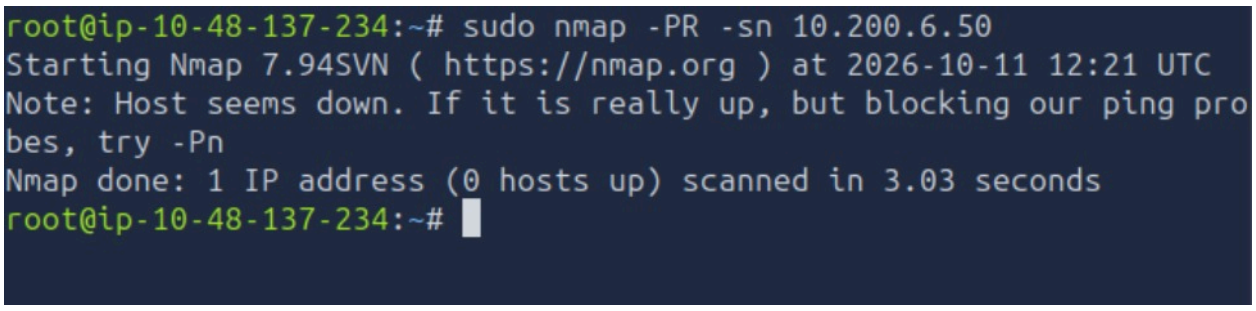
STEP-BY-STEP PROCEDURE & SCREENSHOT PLACEMENT GUIDE

Task 1: Discovering Live Hosts Using ARP Scans

1. Attempt an ARP scan against the target across the router boundary using the command:
2. Bash

```
Shell
sudo nmap -PR -sn 10.200.6.50
```

Observe that Nmap reports 0 hosts up (Host seems down) because ARP packets cannot traverse routers into another subnet.

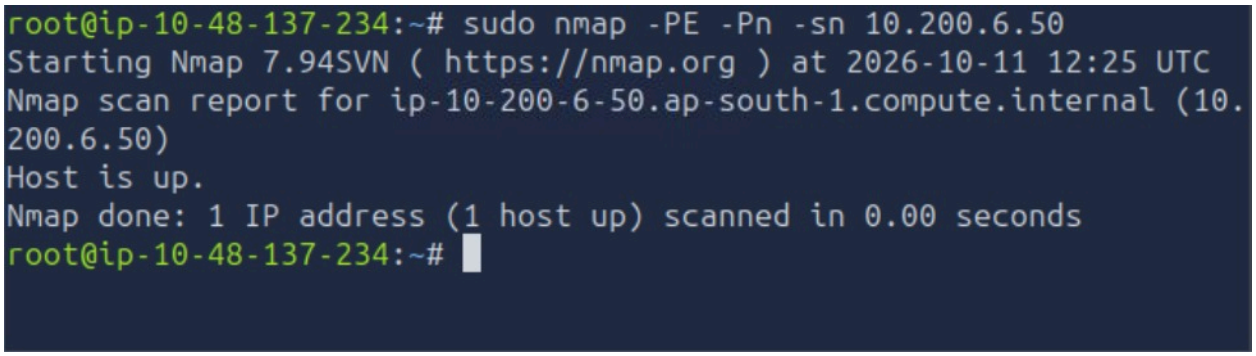


Task 2: Discovering Live Hosts Using ICMP Scans

1. Perform an ICMP Echo Request scan with -Pn enabled to bypass ping blocks:
2. Bash

```
Shell
sudo nmap -PE -Pn -sn 10.200.6.50
```

Observe that Nmap receives a valid response and confirms Host is up.

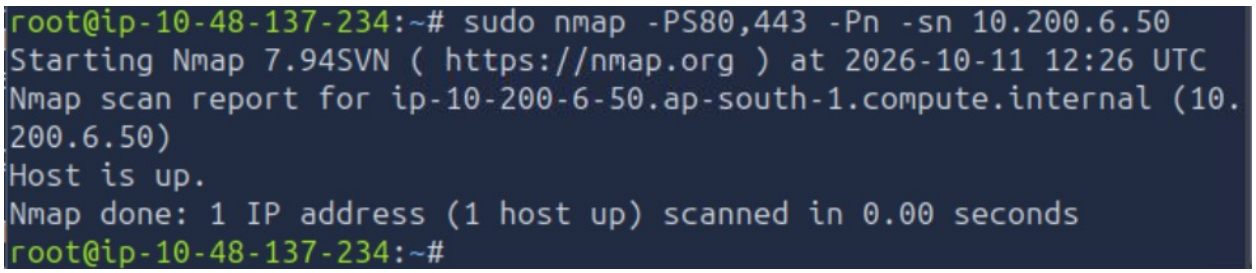


Task 3: Discovering Live Hosts Using TCP Ping Scans

1. Perform a TCP SYN ping scan targeted at ports 80 and 443:
2. Bash

```
Shell
sudo nmap -PS80,443 -Pn -sn 10.200.6.50
```

Observe that Nmap receives TCP responses (SYN/ACK or RST) and confirms Host is up.



Task 4: Discovering Live Hosts Using UDP Ping Scans

1. Perform a UDP ping scan targeted at port 53:
2. Bash

```
Shell
sudo nmap -PU53 -Pn -sn 10.200.6.50
```

Observe that Nmap receives an ICMP Port Unreachable message from the closed port and confirms Host is up.

```
root@ip-10-48-137-234:~# sudo nmap -PU53 -Pn -sn 10.200.6.50
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-11 12:28 UTC
Nmap scan report for ip-10-200-6-50.ap-south-1.compute.internal (10.
200.6.50)
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
root@ip-10-48-137-234:~#
```

OBSERVATION TABLE

Task	Scan Type	Command Executed	Result Displayed	Key Reason / Inference
Task 1	ARP Scan	sudo nmap -PR -sn 10.200.6.50	0 hosts up	ARP packets operate at Layer 2 and cannot cross router subnets.
Task 2	ICMP Scan	sudo nmap -PE -Pn -sn 10.200.6.50	Host is up	Responded to ICMP probe when ping checks were bypassed (-Pn).
Task 3	TCP Ping Scan	sudo nmap -PS80,443 -Pn -sn 10.200.6.50	Host is up	Received TCP response (SYN/ACK or RST) from the target ports.
Task 4	UDP Ping Scan	sudo nmap -PU53 -Pn -sn 10.200.6.50	Host is up	Received ICMP Port Unreachable message from the target.

Result: The experiment to understand the features of Nmap for discovering live hosts using ARP, ICMP, TCP, and UDP scans on the TryHackMe platform has been studied and executed successfully.